INTERNSHIP COMPLETION REPORT

Submitted to

NextGen Technologies

Project Title

COLLEGE STUDENT PLACEMENT PREDICTION SYSTEM:

AI-Powered Placement Prediction and Career Guidance Platform Using Machine Learning

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DECLARATION

I, Muthu Rakshana R (Roll No. E24AI031), student of B.Sc Computer Science with Artificial Intelligence at SDNB Vaishnav College for Women, hereby declare that this Internship Completion Report titled **"College Student Placement Prediction System: AI-Powered Placement Prediction and Career Guidance Platform"** is a genuine record of work carried out by me at GB Tech Corp during the period 12 May 2026 to 12 June 2026, under the supervision of Ms. JayaPriya S.

This report has been prepared exclusively for submission to NextGen Technologies and has not been submitted elsewhere for any academic or professional purpose.

Name: Muthu Rakshana R
Date: 12 June 2026

Signature of Intern: ______________________________
Countersigned by Supervisor: ____________________

CERTIFICATE

This is to certify that Ms. Muthu Rakshana R, B.Sc Computer Science with Artificial Intelligence at SDNB Vaishnav College for Women (Roll No. E24AI031), has successfully completed a one-month internship at GB Tech Corp from 12 May 2026 to 12 June 2026.

During the internship, she worked on the project titled "College Student Placement Prediction System: AI-Powered Placement Prediction and Career Guidance Platform" and demonstrated commendable technical ability, diligence, and professionalism throughout the engagement.

This report is hereby certified as an authentic record of her internship work, submitted for the evaluation of NextGen Technologies.

Supervisor Name: Jayapriya S
Organization: GB Tech Corp
Date: 12 June 2026

Supervisor Signature: ______________________________

ABSTRACT

The College Student Placement Prediction System is a Machine Learning-based web application developed to predict the placement opportunities of students based on their academic performance, technical skills, communication abilities, internship experience, and project involvement.

The system utilizes a Random Forest Classifier model trained on a dataset containing 10,000 student records with 9 attributes. The model analyses various factors affecting employability and predicts whether a student is likely to secure placement opportunities.

In addition to placement prediction, the application provides interactive visualizations, performance analytics, and a Career Guidance Assistant that helps students understand their strengths and areas for improvement.

The project was developed using Python, Streamlit, Scikit-Learn, Pandas, NumPy, Plotly, and Matplotlib. The complete application was managed through GitHub and successfully deployed on Streamlit Cloud for online accessibility.

The developed system achieved a prediction accuracy of 95%, demonstrating the effectiveness of Machine Learning techniques in educational analytics and career planning applications.

Keywords: Machine Learning, Placement Prediction, Random Forest, Student Analytics, Career Guidance, Streamlit, Data Visualization

CHAPTER 1: INTRODUCTION

1.1 Background

Campus placements play a crucial role in shaping the careers of students by providing opportunities to secure employment before graduation. However, assessing a student's placement readiness based on academic performance, technical skills, communication abilities, internships, and project experience can be challenging.

Traditional methods of evaluating placement potential often rely on manual assessments and may not accurately predict placement outcomes. With the advancement of Machine Learning, predictive models can analyse student data and identify patterns associated with successful placements.

GB TECH CORP assigned this internship project to develop a Machine Learning-based College Student Placement Prediction System capable of predicting placement opportunities and providing career guidance. The system aims to help students understand their strengths and areas for improvement while assisting them in preparing effectively for placement drives.

1.2 Objectives

- To develop a Machine Learning-based placement prediction system.
- To analyse student academic and skill-related attributes.
- To predict placement opportunities using Random Forest Classification.
- To visualize student performance through interactive dashboards.
- To provide career guidance and placement preparation suggestions.
- To deploy the application online for easy accessibility.

1.3 Scope

The College Student Placement Prediction System predicts placement opportunities using Machine Learning based on student academic performance, technical skills, communication abilities, internships, and project experience. The system provides placement predictions, interactive visualizations, and career guidance through a Streamlit-based web application deployed on Streamlit Cloud.

1.4 Report Structure

Chapter 2 presents the literature review. Chapter 3 describes the dataset and preprocessing techniques. Chapter 4 explains the system architecture and methodology. Chapter 5 discusses results and evaluation. Chapter 6 covers challenges and learnings. Chapter 7 provides recommendations and future enhancements. The report concludes with references and appendices.

CHAPTER 2: LITERATURE REVIEW

2.1 Machine Learning in Educational Analytics

Educational institutions increasingly use Machine Learning techniques to analyse student performance and predict academic outcomes. Predictive analytics enables institutions to identify at-risk students, improve learning strategies, and enhance placement opportunities.

Machine Learning algorithms such as Decision Trees, Random Forests, Support Vector Machines, and Neural Networks have demonstrated effectiveness in educational prediction tasks.

2.2 Placement Prediction Systems

Placement prediction systems aim to estimate the likelihood of students securing employment opportunities based on historical and current performance indicators.

Previous studies have shown that factors such as CGPA, internships, projects, communication skills, and technical knowledge significantly influence placement outcomes.

2.3 Career Guidance Systems

Modern career guidance platforms assist students in identifying skill gaps and preparing for future opportunities. Combining predictive analytics with guidance systems enhances the overall effectiveness of student career planning.

2.4 Research Gap

Many existing placement prediction systems focus solely on prediction and do not provide integrated visualization dashboards, career guidance, and cloud accessibility.

This project addresses these limitations by integrating prediction, analytics, visualization, and guidance into a single web-based platform.

CHAPTER 3: DATASET AND PREPROCESSING

3.1 Dataset Description

The dataset used for this project consists of student placement-related information collected for Machine Learning model training and evaluation. The dataset contains academic, technical, and skill-based attributes that influence placement outcomes.

Attribute	Details
Total Records	10,000
Number of Features	9
Target Variable	Placement Status
Dataset Type	Structured Tabular Data
Algorithm Used	Random Forest Classifier
Prediction Type	Binary Classification
Accuracy Achieved	95%

The dataset includes important student attributes such as CGPA, IQ Score, Communication Skills, Internship Experience, Technical Skills, Project Experience, and other performance indicators used to predict placement opportunities.

3.2 Data Preprocessing

The dataset was preprocessed to improve data quality and ensure effective model training.

- **Data Cleaning:** Removed inconsistencies and validated dataset entries.
- **Missing Value Handling:** Checked for missing values and ensured data completeness.
- **Feature Selection**: Selected relevant attributes that significantly impact placement outcomes.
- **Data Transformation:** Converted data into a suitable format for Machine Learning processing.
- **Data Splitting:** Divided the dataset into training and testing datasets for model evaluation.
- **Model Preparation:** Prepared the processed data for Random Forest Classifier training and prediction.

These preprocessing steps improved model performance and contributed to achieving high prediction accuracy.

3.3 Exploratory Analysis

The analysis included:

- Placement Distribution Analysis.
- CGPA and Placement Correlation.
- Communication Skill Analysis.
- Internship Experience Analysis.
- Technical Skill Evaluation.
- Feature Correlation Analysis.

The visualizations generated during EDA helped identify patterns influencing placement outcomes and supported the selection of relevant features for model development.
The analysis confirmed that academic performance, communication skills, internships, and technical competencies play a significant role in determining student placement opportunities.

CHAPTER 4: MODEL ARCHITECTURE AND METHODOLOGY

4.1 System Architecture Overview

The College Student Placement Prediction System operates as a Machine Learning-based web application that predicts placement opportunities based on student academic and skill-related information.

The system follows the workflow below:

- **Stage 1 — Student Data Collection:** Students enter their academic and skill-related information through the Streamlit application.
- **Stage 2 — Data Processing:** The entered information is validated and preprocessed for prediction.
- **Stage 3 — Placement Prediction:** The trained Random Forest Classifier analyses the input data and predicts placement outcomes.
- **Stage 4 — Result Visualization:** Prediction results and analytical insights are displayed through interactive dashboards and charts.
- **Stage 5 — Career Guidance:** Students receive placement preparation suggestions and career guidance based on their profile.

The complete system provides a user-friendly interface for prediction, analysis, and career support.

4.2 Machine Learning Model: Random Forest Classifier

The Random Forest Classifier was selected as the primary Machine Learning algorithm due to its high accuracy, robustness, and ability to handle multiple features effectively.

Advantages of Random Forest include:

- High prediction accuracy.
- Reduced risk of overfitting.
- Ability to process large datasets efficiently.
- Effective handling of both numerical and categorical data.
- Strong performance in classification problems.

The model was trained using a dataset containing 10,000 student records and achieved a prediction accuracy of 95%.

Model Components

Component	Description
Input Data	Student Academic and Skill Attributes
Algorithm	Random Forest Classifier
Dataset Size	10,000 Records

Features	9 Attributes
Output	Placement Prediction
Prediction Type	Binary Classification
Accuracy	95%

4.3 Training Configuration

The Random Forest model was trained using the following configuration:

Parameter	**Value**
Algorithm	Random Forest Classifier
Dataset Size	10,000 Records
Features	9
Prediction Type	Binary Classification
Accuracy Achieved	95%
Programming Language	Python
Machine Learning Library	Scikit-Learn

The model was trained and evaluated using processed student data to ensure reliable placement prediction performance.

4.4 Streamlit Application Development

The application was developed using Streamlit to provide an interactive and responsive user interface.

The application includes the following modules:

- **Home Page:** Displays project information, workflow, and platform features.
- **Student Details Module:** Collects student information required for prediction.
- **Placement Prediction Module:** Generates placement predictions using the trained Random Forest model.
- **Visualization Dashboard:** Displays interactive charts and analytics related to student performance.
- **Career Guidance Assistant:** Provides placement preparation guidance and career recommendations.

- **Placement Awareness Module:** Contains informations and details about placement ,and developer information.

The Streamlit framework enabled rapid development, attractive UI design, and seamless integration with Machine Learning models.

4.5 Deployment Architecture

The project follows a cloud-based deployment architecture:

Student/User
↓
Streamlit Web Application
↓
Machine Learning Prediction Engine
↓
Prediction Results
↓
Visualization Dashboard
↓
Career Guidance Assistant

The application source code is managed through GitHub and deployed online using Streamlit Cloud.

Deployment URL:
https://machinelearningproject-internship.streamlit.app/

The deployed system allows users to access placement prediction and career guidance services from any device with internet connectivity.

CHAPTER 5: RESULTS AND EVALUATION

5.1 Model Training Results

The Random Forest Classifier was successfully trained using a dataset containing 10,000 student records and 9 features related to academic performance, technical skills, communication abilities, internship experience, and project work.

The model learned patterns associated with successful student placements and generated reliable classification results.

The training process resulted in a highly accurate predictive model with consistent performance across different student profiles.

5.2 Prediction Performance

The trained model was evaluated using unseen testing data to measure its prediction capability.

Performance Metrics

Metric	Value
Accuracy	95%
Precision	94%
Recall	95%
F1-Score	95%

The model achieved strong classification performance, indicating its ability to accurately predict student placement outcomes based on the provided attributes.

The high accuracy confirms that factors such as CGPA, communication skills, internships, technical skills, and project experience significantly influence placement success.

5.3 Prediction Results

The system generates predictions based on student information entered through the web application.

Possible Outcomes

Prediction Result	Description
Placed	Student is likely to secure placement opportunities
Not Placed	Student requires improvement in certain areas

The prediction results are displayed instantly along with supporting insights and recommendations.

5.4 Visualization Dashboard Results

An interactive visualization dashboard was developed using Streamlit and Plotly to provide meaningful insights into student placement trends and career readiness factors. The dashboard enables users to analyse various parameters that influence placement outcomes through easy-to-understand graphical representations.

Implemented visualizations include:

- **Placement Distribution (Pie Chart):** Displays the percentage of placed and non-placed students in the dataset.
- **CGPA Analysis (Bar Chart):** Shows the academic performance of students and its impact on placement opportunities.
- **IQ Score Distribution (Histogram):** Illustrates the distribution of student IQ scores and overall aptitude levels.
- **Communication Skills Analysis (Scatter Plot):** Highlights the relationship between communication skills and placement outcomes.
- **CGPA Trend Analysis (Line Chart):** Visualizes academic performance trends across students.
- **Internship Experience Analysis (Grouped Bar Chart):** Demonstrates how internship experience contributes to placement success.

The dashboard provides an intuitive and interactive environment for analyzing placement-related factors. These visualizations help students, faculty members, and placement coordinators understand key performance indicators and make data-driven decisions to improve placement readiness.

5.5 Career Guidance Assistant Evaluation

The Career Guidance Assistant module was successfully integrated into the application.

The module provides guidance related to:

- Placement preparation.
- Internship importance.
- Communication skill improvement.
- Project development recommendations.
- Software development career guidance.
- Machine Learning career guidance.

The assistant enhances student engagement and provides practical career development support.

5.6 User Interface Evaluation

The Streamlit-based application was evaluated for usability, responsiveness, and user experience.

Evaluation Results

Feature	Status
Login System	Successfully Working
Home Page Navigation	Successfully Working
Student Details Module	Successfully Working
Prediction Module	Successfully Working

Visualization Dashboard	Successfully Working
Career Guidance Assistant	Successfully Working

The user interface provides smooth navigation, attractive design, and efficient interaction.

5.7 Deployment Results

The application was successfully deployed using Streamlit Cloud and connected to GitHub for version control and project management.

Deployment Details

Parameter	Value
Hosting Platform	Streamlit Cloud
Source Control	GitHub
Application Status	Live
Accessibility	Online
Deployment Status	Successful

Deployment URL:
https://machinelearningproject-internship.streamlit.app/

The deployed application allows users to access placement prediction and career guidance services through a web browser without requiring software installation.

5.8 Overall Evaluation

The College Student Placement Prediction System successfully achieved all project objectives.

Major accomplishments include:

- Development of a Machine Learning-based placement prediction model.
- Achievement of 95% prediction accuracy.
- Creation of an interactive Streamlit web application.
- Integration of visualization dashboards.
- Development of a Career Guidance Assistant.
- Successful GitHub integration and cloud deployment.

The project demonstrates the practical application of Machine Learning in educational analytics and career guidance systems while providing a user-friendly platform for students.

CHAPTER 6: CHALLENGES AND LEARNINGS

6.1 Challenges Encountered

Challenge	Solution Applied	Outcome
Data preprocessing and feature selection	Performed data cleaning, validation, and selected relevant placement-related features	Improved model performance and prediction reliability
Achieving high prediction accuracy	Experimented with model parameters and optimized the Random Forest Classifier	Achieved 95% prediction accuracy
Designing an attractive user interface	Implemented custom CSS styling and responsive layouts in Streamlit	Enhanced user experience and visual appeal
Multi-page application development	Structured the project into separate modules and managed page navigation	Smooth navigation across all application pages
Career Guidance Assistant integration	Developed an interactive guidance module with placement-related recommendations	Improved student engagement and career support functionality
GitHub repository management and deployment	Organized project files, configured dependencies, and integrated GitHub with Streamlit Cloud	Successful cloud deployment and online accessibility

6.2 Technical Learnings

- Gained practical experience in Machine Learning model development using the Random Forest Classifier.
- Learned data preprocessing, feature selection, and dataset analysis techniques.
- Improved knowledge of Exploratory Data Analysis (EDA) and data visualization.
- Developed skills in building interactive web applications using Streamlit.
- Learned custom UI development using HTML and CSS within Streamlit applications.
- Gained experience in GitHub repository management and version control practices.
- Successfully deployed a Machine Learning application using Streamlit Cloud.
- Learned how to integrate Machine Learning models with user-friendly web interfaces.

6.3 Professional Learnings

- Improved project planning and task management skills.
- Learned to document project progress and prepare technical reports.
- Enhanced problem-solving and debugging abilities during application development.
- Improved communication skills through project presentations and reporting.
- Developed the ability to learn and implement new technologies independently.
- Gained practical exposure to real-world Machine Learning project development workflows.
- Understood the complete lifecycle of a Machine Learning project from development to deployment.

CHAPTER 7: RECOMMENDATIONS AND FUTURE WORK

7.1 Recommendations for Future Development

• Enhance the Career Guidance Assistant by integrating AI-powered conversational support for personalized career recommendations.
• Implement a Resume Analysis module to evaluate student resumes and provide improvement suggestions.
• Add a Mock Interview Assistant to help students practice technical and HR interview questions.
• Integrate skill gap analysis to identify areas where students need improvement before placement drives.
• Develop a company recommendation system based on student skills, interests, and academic performance.
• Store prediction history and student records in a database for long-term tracking and analysis.
• Improve dashboard functionality with advanced analytics and interactive reporting features.
• Implement user authentication and role-based access for students, faculty members, and administrators.

7.2 Future Technical Improvements

• Experiment with advanced Machine Learning algorithms such as XGBoost, Gradient Boosting, and Neural Networks for improved prediction accuracy.
• Integrate Large Language Models (LLMs) to provide intelligent career counseling and placement preparation guidance.
• Develop a mobile-responsive version of the application for improved accessibility.
• Implement cloud database integration for secure data storage and retrieval.
• Add real-time placement trend analysis and reporting capabilities.
• Enable automatic model retraining when new student data becomes available.
• Integrate email notification systems to share prediction reports and career guidance recommendations with students.
• Expand the system to support internship prediction and career pathway analysis.

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APPENDIX A — DELIVERABLES CHECKLIST

Deliverable	Description	Status
Dataset Preparation and Preprocessing	Student placement dataset preparation and preprocessing	Completed
Machine Learning Model Development	Random Forest Classifier training and evaluation	Completed
Placement Prediction Module	Student placement prediction functionality	Completed
Visualization Dashboard	Interactive charts and analytics dashboard	Completed
Career Guidance Assistant	Placement preparation and career guidance module	Completed
Streamlit Web Application	Multi-page web application development	Completed
GitHub Repository	Project source code management and version control	Completed
Streamlit Cloud Deployment	Online application deployment and hosting	Completed
Project Documentation	Technical documentation and project report	Completed
Internship Project Report	Final internship report submission	Completed

APPENDIX B — ABBREVIATIONS

Abbreviation	Full Form
ML	Machine Learning
AI	Artificial Intelligence
RF	Random Forest
EDA	Exploratory Data Analysis
UI	User Interface
UX	User Experience
CSV	Comma Separated Values
API	Application Programming Interface
URL	Uniform Resource Locator
IDE	Integrated Development Environment
SQL	Structured Query Language
Git	Distributed Version Control System
KPI	Key Performance Indicator
LLM	Large Language Model
CGPA	Cumulative Grade Point Average